Research Notes on the Riemann Hypothesis

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Résumé

Ce document constitue un cadre de recherche, de vérification et de documentation relatif à l'hypothèse de Riemann.

Aucune démonstration de l'hypothèse de Riemann n'est revendiquée dans ce manuscrit.

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1 Statement of the Problem

Définition 1.1. For $\Re(s) > 1$,

$$\zeta(s) = \sum_{n=1}^{\infty} \frac{1}{n^s}.$$

The function admits an analytic continuation to the entire complex plane except for a simple pole at $s = 1$.

Théorème 1.2 (Riemann Hypothesis)**.** *Every non-trivial zero ρ of $\zeta(s)$ satisfies*

$$\Re(\rho) = \frac{1}{2}.$$

The vertical line

$$\Re(s) = \frac{1}{2}$$

is called the critical line.
The domain

$$0 < \Re(s) < 1$$

is called the critical strip.

2 Known Results

The following results are established.

2.1 Euler Product

For $\Re(s) > 1$,

$$\zeta(s) = \prod_{p} \left(1 - p^{-s}\right)^{-1}.$$

This identity links the zeta function to prime numbers.

2.2 Functional Equation

The zeta function satisfies

$$\zeta(s) = 2^s \pi^{s-1} \sin\left(\frac{\pi s}{2}\right) \Gamma(1-s)\zeta(1-s).$$

2.3 Prime Number Theorem

The prime-counting function satisfies

$$\pi(x) \sim \frac{x}{\log x}.$$

2.4 Hardy Theorem

There exist infinitely many zeros on the critical line.

2.5 Computational Evidence

The first very large collections of zeros that have been checked numerically lie on the critical line.

These verifications remain finite and therefore do not constitute a proof.

3 Candidate Research Strategies

This section collects possible directions of investigation.

3.1 Equivalent Criteria

Several statements are known to be equivalent to RH.

Examples include :

— Li Criterion.

— Weil Criterion.

— Nyman-Beurling Criterion.

— Báez-Duarte Criterion.

A possible strategy is to simplify one of these formulations.

3.2 Spectral Approaches

The Hilbert–Pólya philosophy suggests that zeros could correspond to eigenvalues of a self-adjoint operator.

Finding such an operator remains open.

3.3 Positivity Approaches

Many equivalent criteria can be reformulated as positivity conditions.
The objective is to identify structures admitting rigorous verification.

3.4 Formal Verification

Incremental formal verification may reduce logical errors and hidden assumptions.

4 Lemma Registry

This section contains only verified intermediate results.

Lemme 4.1 (Template)**.** *Let A be a mathematical object satisfying hypotheses H.*

Then property P holds.

Démonstration. Proof placeholder.

The proof must explicitly justify all limits, summations, integrations, and analytic continuations.

No proof is provided at this stage. □

4.1 Open Lemmas

The following identifiers are reserved :
— LEM-0001
— LEM-0002
— LEM-0003
Their statements will be added as the project develops.

5 Failure Analysis

Many alleged proofs of RH fail for recurring reasons.

5.1 Circular Reasoning

An argument may unknowingly assume a result equivalent to RH.

5.2 Invalid Exchanges

Common issues involve unjustified exchanges between :

$$\lim \quad \text{and} \quad \sum$$

or

$$\int \quad \text{and} \quad \sum.$$

5.3 Uniformity Problems

An estimate valid for fixed parameters may fail under a limit process.

5.4 Analytic Continuation Issues

Extending a function beyond its domain of convergence requires full justification.

5.5 Numerical Overreach

Finite numerical verification is not equivalent to a proof of an infinite statement.

6 Computational Evidence

Computations play an important role in research.

6.1 Objectives

— Validate software implementations.
— Explore numerical behaviour.
— Detect counterexamples.
— Generate conjectures.

6.2 Limitations

No finite computation can prove RH.
Computations only provide information about finitely many cases.

6.3 Experiment Registry

Current experiments include :
— EXP-001 - Zeta Sanity Checks.
Further experiments may investigate :
— Gram points.
— Hardy Z-function.
— Li coefficients.
— Jensen polynomials.
— Random matrix theory.

7 Conclusion

The Riemann Hypothesis remains one of the central unsolved problems of mathematics.

This repository is intended as :

— a research framework ;
— a reproducible computational environment ;
— a documentation platform ;
— a proof-management system ;
— a preparation ground for formal verification.

Future work will focus on :

1. equivalent formulations ;
2. spectral approaches ;
3. formal verification ;
4. reproducible numerical experiments.

At the current stage no proof of RH is claimed.

Références